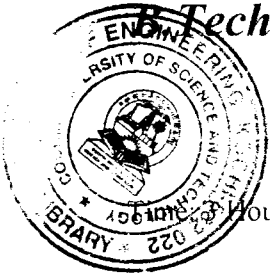


--	--	--	--	--	--	--	--



B.Tech. Degree VII Semester Examination November 2018

ME 15-1702 VIBRATION AND NOISE CONTROL (2015 Scheme)

Maximum Marks: 60

PART A (Answer ALL questions)

(10 × 2 = 20)

- I. (a) In what ways is the response of a system with Coulomb damping different from that of systems with other types of damping?
- (b) Is the frequency of a damped free vibration smaller or greater than the natural frequency of the system? Explain.
- (c) Define the term magnification factor. How is the magnification factor related to the frequency ratio?
- (d) A stepped shaft has the following dimensions: $L_1=0.5$ m, $L_2 = 0.3$ m; $d_1 = 0.015$ m, $d_2 = 0.012$ m. Calculate the equivalent length of the shaft whose diameter is same as d_1 .
- (e) What is the significance of eigen values and eigen vectors in a multi degree of freedom vibrating system? Explain.
- (f) Explain coordinate coupling by considering a simplified two degree of freedom vibrating system.
- (g) Derive the wave equation by considering a uniform rod subjected to axial force causing longitudinal vibrations.
- (h) Differentiate between Dunkerley's and Rayleigh's methods in determining the frequency of transverse vibrations.
- (i) Obtain a relation between sound power level and sound intensity level.
- (j) Determine the overall sound pressure level when two machines generating 85 dB each operate simultaneously.

PART B

(4 × 10 = 40)

- II. Consider a spring mass system, with $k=4000$ N/m and $m = 10$ kg, subject to a harmonic force $F(t) = 400.\cos(30t)$ N. Find and plot the total response of the system under the following initial conditions: (10)
 - (i) $x_0 = 0.1$ m and $v_0 = 0$
 - (ii) $x_0 = 0.1$ m and $v_0 = 10$ m/s

OR

- III. A metal block, placed on a rough surface, is attached to a spring and is given an initial displacement of 10 cm from its equilibrium position. After 5 cycles of oscillation in 2 seconds, the final position of the metal block is found to be 1 cm from its equilibrium position. Determine the coefficient of friction between the surface and the metal block. (10)

(P.T.O.)